

Program9: Organize Kubernetes pods using labels and namespace

Project Structure:

Pgm9

->namespaces.yaml

->pods.yaml

Step1: Create pods.yaml

```
GNU nano 7.2 namespace.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

Step2: create namespace.yaml

```
GNU nano 7.2 pods.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 3: Apply namespace and pods

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg9$ kubectl apply -f pods.yaml
namespace/dev created
namespace/prod created
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg9$ kubectl apply -f namespace.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

Step 4: Verify pods in each namespace

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           41s
frontend-pod  1/1     Running   0           41s
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           44s
frontend-pod  1/1     Running   0           44s
```

Step 5: Filter pods by label

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg9$ kubectl get pods -n prod -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           2m30s
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           2m37s
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           2m42s
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           2m49s
```